



Cambridge (CIE) IGCSE Computer Science



Your notes

Input & Output Devices

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Your notes

Input Devices

Input Devices

What is an Input Device?

- Input devices are **hardware components** that **allow users to interact** with a computer system
- They enable the user to **input data or commands into the system**, which the computer then processes to produce an output

Types of input devices

Device	What It Does	Where It Is Used	Why It Is Used
Barcode Scanner	Reads barcodes to capture product information.	- Supermarkets - Warehouses - Libraries	Quickly and accurately retrieves product details for stock and checkout
Digital Camera	Captures photographs or videos as digital images	- Photography - Smartphones - Webcams	High-quality visual content
Keyboard	Inputs text and commands by pressing keys	- Computers - Tablets - Laptops	- Type documents - Enter data - Control software
Microphone	Captures analogue sound and converts it into digital signals	- Recording studios - Smartphones - Computers	- Audio recording - Voice communication



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Optical Mouse	Moves the cursor on the screen based on hand movements over a surface	- Computers - Laptops	Precise control and navigation on a computer screen
QR Code Scanner	Reads QR codes to access stored information	- Retail shops - Mobile apps	- Quickly retrieves data from QR codes - Website links or product information
Touch Screen (Resistive)	Detects touch by pressing a flexible screen layer	- ATMs - Older smartphones and tablets	Direct interaction with the device
Touch Screen (Infra-red)	Detects touch by interrupting infrared light beams across the screen	- Kiosks - Large touch displays	- Accurate touch detection - Works well in various lighting conditions
Touch Screen (Capacitive)	Detects touch through changes in electrical charge on the screen surface	- Modern smartphones - Tablets - Touch monitors	Smooth, responsive touch experience with multi-touch capabilities
Two-Dimensional (2D) Scanner	Captures flat, 2D images of documents or photos	- Offices - Libraries - Schools	Digitises documents and photos for storage, editing, or sharing
Three-Dimensional (3D) Scanner	Captures detailed 3D shapes and dimensions of objects	- Manufacturing - Design - Medical imaging	Creates digital models of objects for design, analysis, or reproduction



Your notes

Output Devices

Output Devices

What is an output device?

- Output devices are **hardware** components that **receive information** from a computer system and **present it to the user** in an understandable form
- They enable the computer to **communicate the results of processed data** or **commands**

Types of output devices

Name of device	What does it do?	Where is it used?	Why is it used?
Actuator	- Converts energy into motion - Makes things move - Performs physical tasks based on instructions from a computer or control system	- Make a robot's arm move - Open a valve - Rotate a motor	Essential in any system that requires movement
Digital Light Processing (DLP) projector	- Projects images onto a screen - Shows images or video by shining light through tiny mirrors	- Cinemas - Classrooms - Large presentations	High-quality images
Inkjet printer	Prints text and images onto paper by spraying tiny droplets of ink	- Home - Schools - Offices	Low cost compared to other types of printer
Laser printer	Heat from a laser fuses toner (fine powder) onto paper	- Offices - Schools	- High-quality prints - Fast print speed



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Light Emitting Diode (LED) screen	Displays images and video using tiny lights called LEDs	- Televisions - Computer monitors - Smartphones	- High-quality images - Bright - Energy efficient - Thin
Liquid Crystal Display (LCD) projector	Projects images or video onto a screen by shining light through liquid crystal panels	- Offices - Classrooms - Home cinemas	- Bright - Excellent balance of quality and affordability
Liquid Crystal Display (LCD) screen	Displays images and video using liquid crystals with a backlight	- Televisions - Monitors - Smartphones - Laptops	- Energy efficient - Thin - Sharp images - Lightweight
Speaker	Converts electrical signals into analogue sound waves to play music	- Televisions - Radios - Smartphones	Multiple shapes and sizes
3d printer	Adds thin layers of plastic, one on top of another to form a three dimensional object	- Manufacturing - Medicine - Education - Home	- Create prototypes - Custom parts

Choosing the right device

When recommending a device for a specific situation, consider the following factors:

- User Needs
 - What tasks will the user be performing? A graphic designer might need a graphics tablet, while a data entry clerk might need a keyboard with a number pad
- User Skills

- Is the user comfortable with the device? A touch screen might be more intuitive for some users, while others might prefer a mouse and keyboard
- Environment
 - Where will the device be used? A wireless mouse might be suitable for a clutter-free office, while a wired mouse might be better for a public computer lab to prevent theft
- Cost
 - Higher-end devices often have more features but are also more expensive. Consider the budget and whether the extra features are worth the cost



Your notes



Your notes

Sensors

Sensors

- Sensors are **input devices**
- They **measure a physical property of their environment** such as light levels, temperature or movement
- Sensors can be used for **monitoring systems and control systems**
 - A monitoring system **tracks the state of a system**, it gathers data and may issue **warning messages**
 - A control system will **control the system** based upon the input from sensors
 - For example, if the water temperature in a fish tank fell below the acceptable level, a control system would **start up a heater**
 - The system will then continue to check the **water temperature readings (outputs)** and when they are within the acceptable range it will **switch off the heater**
- A process where **outputs are recycled and used as inputs**, creating a continuous cycle is called a **feedback loop**



Examiner Tips and Tricks

- You are likely to be given a scenario and asked to identify the sensors that would be used.
- The 14 sensors you need to be aware of are in the table below

Sensor type	What it measures	Typical use
Acoustic	Sound levels	To detect changes in sound levels of industrial machinery To monitor noise pollution In security system to detect suspicious sounds
Accelerometer	Acceleration rate, tilt, vibration	Detecting sudden changes in vehicle movement and deploy safety features if needed In mobile phones to detect orientation of the device



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Flow	Rate of gas, liquid or powder flow	Detect changes in the flow through pipes in water system
Gas	Presence of a gas e.g. carbon monoxide	Detect levels of gas in confined spaces Detect gas levels when fixing gas leaks
Humidity	Levels of water vapour	Monitor humidity in greenhouses
Infra-red	Detecting motion or a heat source	Security systems detecting intruders who break the beam Measures heat radiation of objects - used by emergency services to detect people
Level	Liquid levels	Detects levels of petrol in a car tank Detect levels of water in a water tank Detect a drop in water levels due to a leak
Light	Light levels	Automatically switching on lights when it gets dark (street lights, headlights)
Magnetic field	Presence and strength	Anti-lock braking system Monitoring rotating machinery such as turbines
Moisture	Presence and levels of moisture	Monitoring moisture in the soil Monitoring dampness in buildings
pH	Acidity or alkaline	Monitoring soil to ensure optimum growing conditions Monitor pH levels in chemical processes
Pressure	Gas, liquid or physical pressure	Monitoring tyre pressure Monitoring pressure in pipes during the manufacturing process
Proximity	Distance	Monitoring the position of objects in robotics



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		Used in safety systems to prevent objects from colliding
Temperature	Temperature in Celsius, Fahrenheit or Kelvin	Used to maintain temperature in swimming pools Used to control temperature in chemical processes



Worked Example

What type of sensor would be used to help drivers when reversing? [1]

A	Motion
B	Infra-red
C	Light
D	Proximity

Is this a monitoring system or a control system? [1]

A	Monitoring
B	Control

Answer

A	Motion
B	Infra-red
C	Light
D	Proximity

A	Monitoring
B	Control